

Smart EV Assistance and Support System

Synopsis

The Smart EV Assistance and Support System is designed to improve the safety, convenience, and maintenance experience of Electric Vehicle (EV) users. It integrates features like an SOS button for emergencies, chatbox for real-time support, charging station locator, service center management, spare parts availability, and a reward system for promoting eco-friendly driving habits.

The system helps EV users receive quick roadside assistance, find nearby charging or service centers, order genuine spare parts, and stay motivated through a digital reward mechanism.

Modules

- User Module – Handles user registration, login, and vehicle information.
- SOS Module – Sends emergency alerts with location to nearby service centers or emergency contacts.
- Chatbox Module – Provides real-time communication with AI support or service representatives.
- Charging Station Module – Displays nearby charging stations with live location and availability.
- Service Center Module – Enables booking and tracking of vehicle servicing.
- Spare Parts Module – Displays and manages available parts from authorized vendors.
- Reward Module – Tracks eco-driving performance and provides redeemable points.
- Admin Module – Manages users, services, inventory, and SOS alerts.

Hardware Requirements

- IoT Device (ESP32 / Raspberry Pi) for SOS integration
- GPS Module for live location tracking
- GSM Module for communication (optional)
- Power Supply Unit
- Smartphone or Computer (User Interface Access)

Software Requirements

- Frontend: HTML, CSS, JavaScript, React
- Backend: Node.js / Python (Flask or Django)
- Database: MySQL / Firebase
- APIs: Google Maps API (for location), Twilio / Firebase (for chat & alerts)
- IDE/Tools: VS Code, XAMPP / Postman, GitHub
- Operating System: Windows / Linux / Android

Expected Outcome

The system provides:

- Instant emergency response through SOS alerts.
- Seamless communication via chatbox support.
- Quick access to charging and service centers.
- Transparent spare parts tracking and maintenance.
- Motivation through rewards, encouraging efficient driving habits.

This project promotes smart, safe, and sustainable EV usage by combining technology, safety, and user engagement in a single platform.